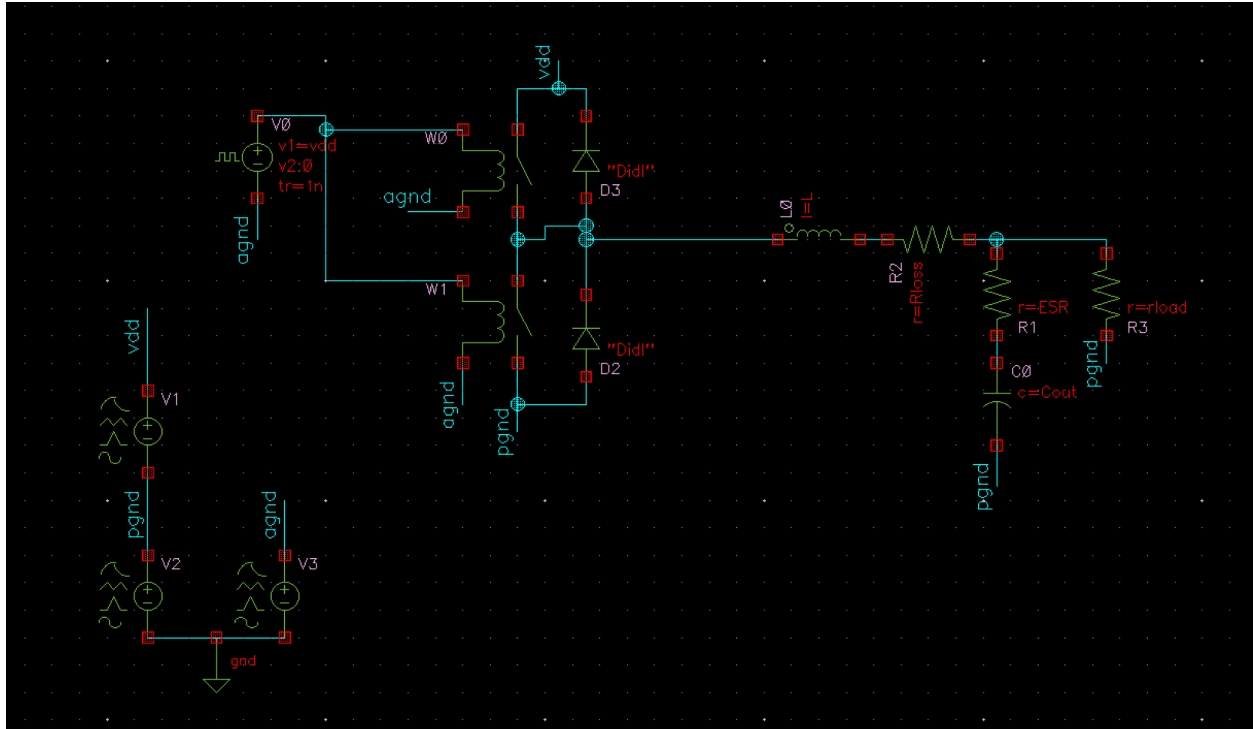


EE 660 Power Management IC Design

Design of a closed-loop voltage-mode buck converter

SAHASRITH BOOTLA (23110064)

SCHMEATIC



ADL

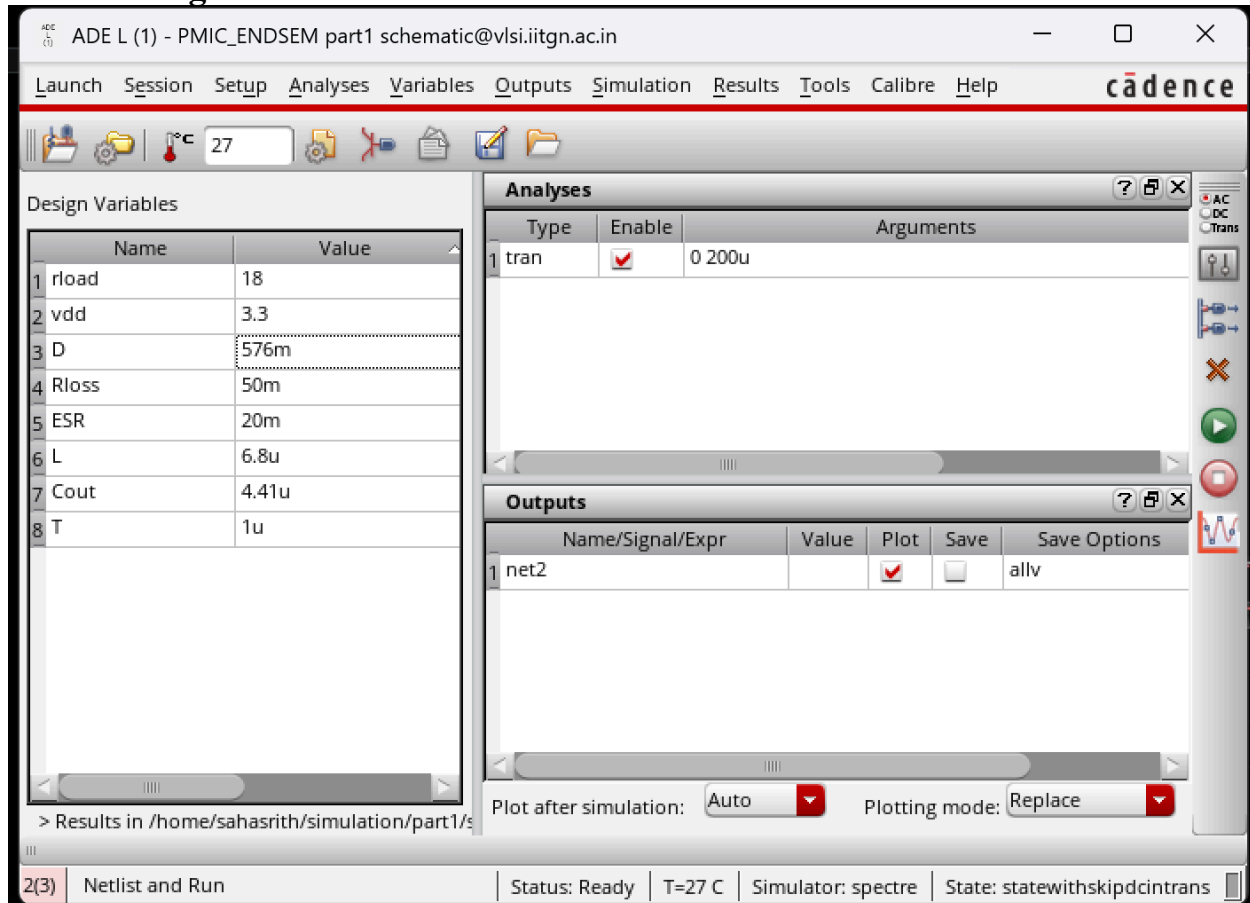
Design Variable	Value
D	550 m
ESR	20 mΩ
L	6.8 μH
Rload	18 Ω
Rloss	50 mΩ
T	1 μs
Cout	4.41 μF
Vdd	3.3 V

Design Variables

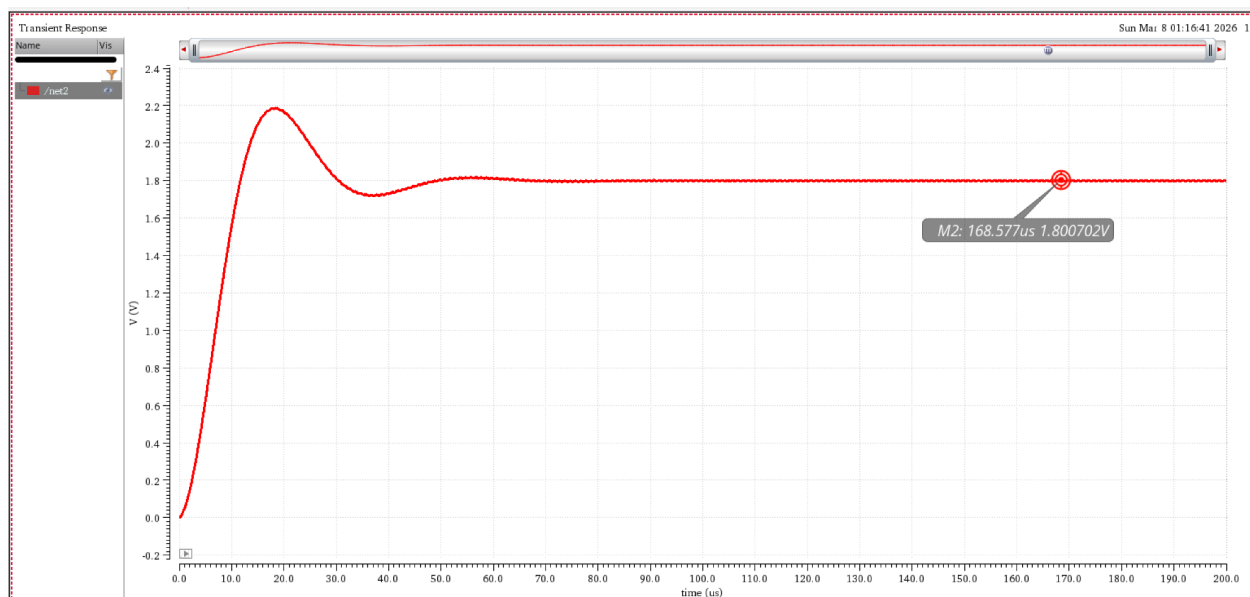
Name	
1 D	550m
2 ESR	20m
3 L	6.8u
4 rload	18
5 Rloss	50m
6 T	1u
7 Cout	4.41u
8 vdd	3.3

PART 1A:

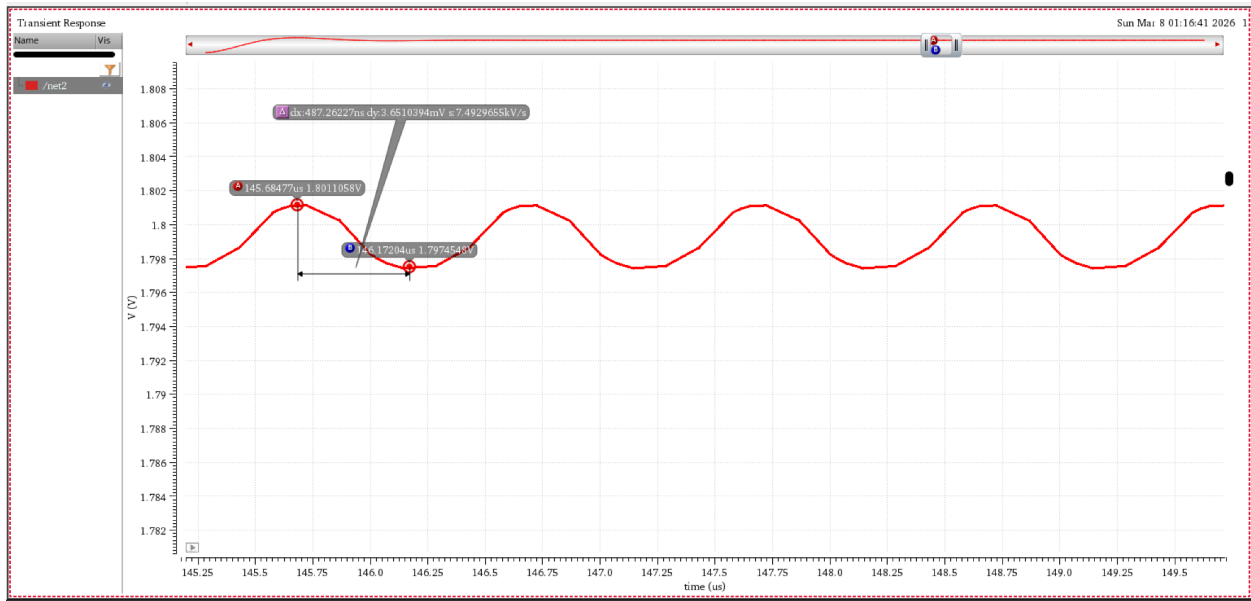
1. For light load



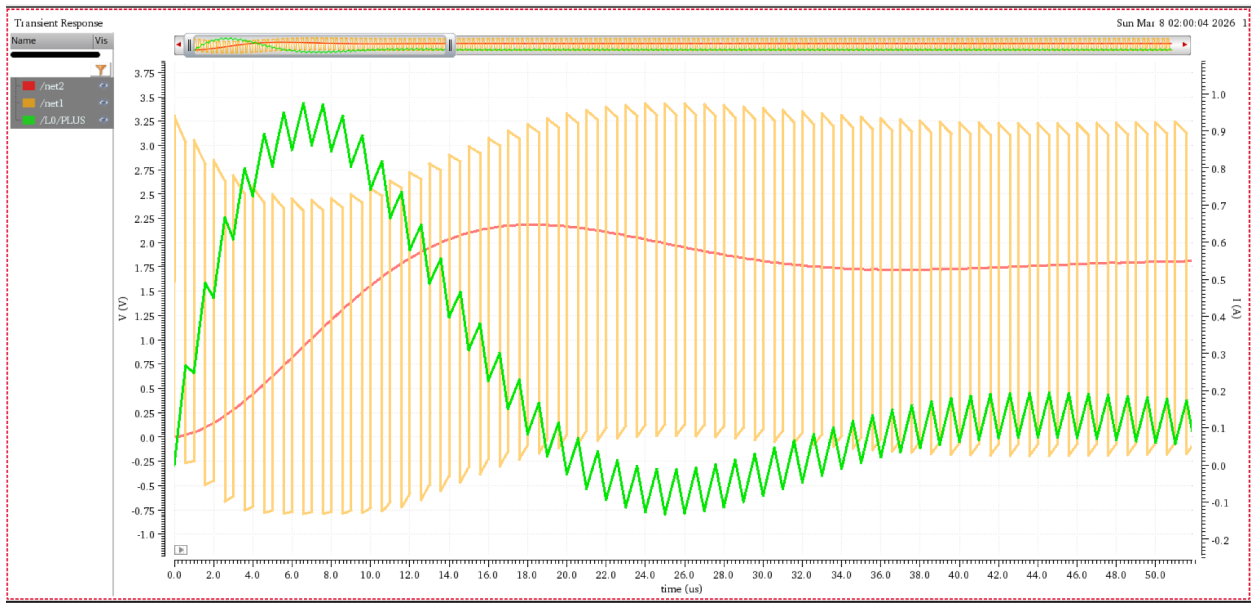
OUTPUT VOLTAGE



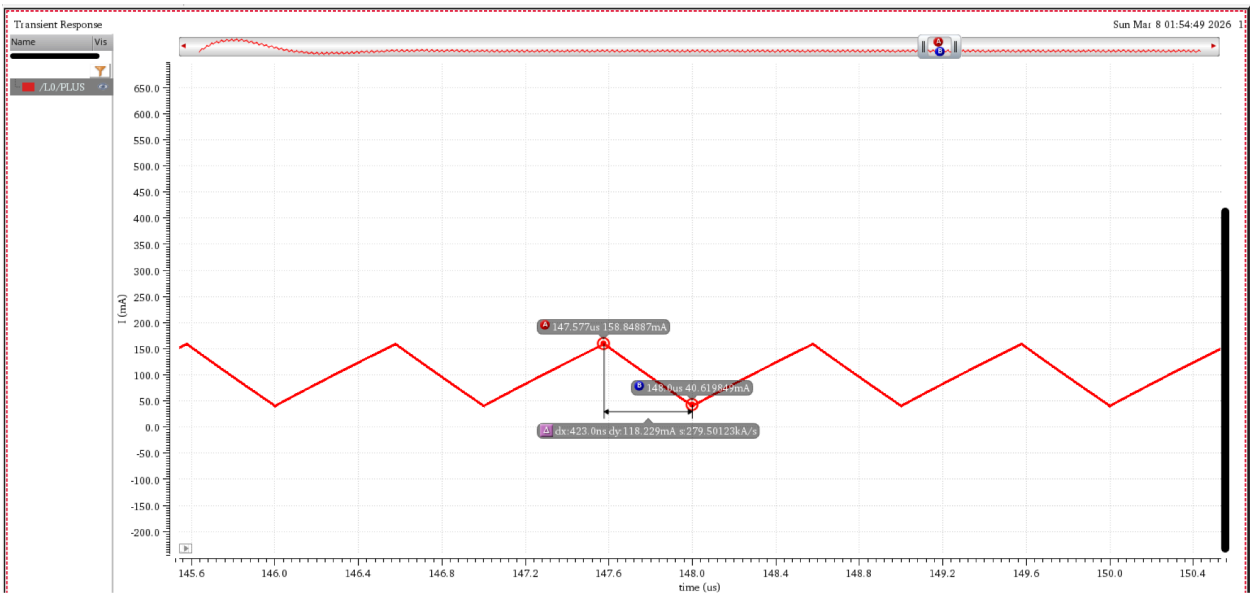
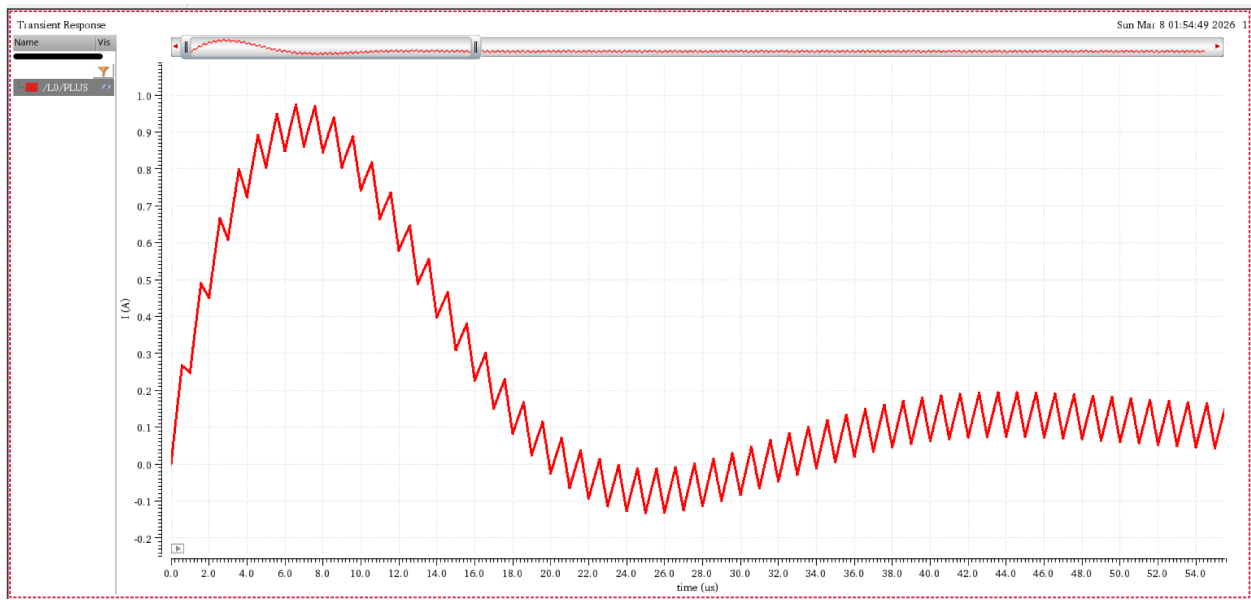
VOLTAGE RIPPLE



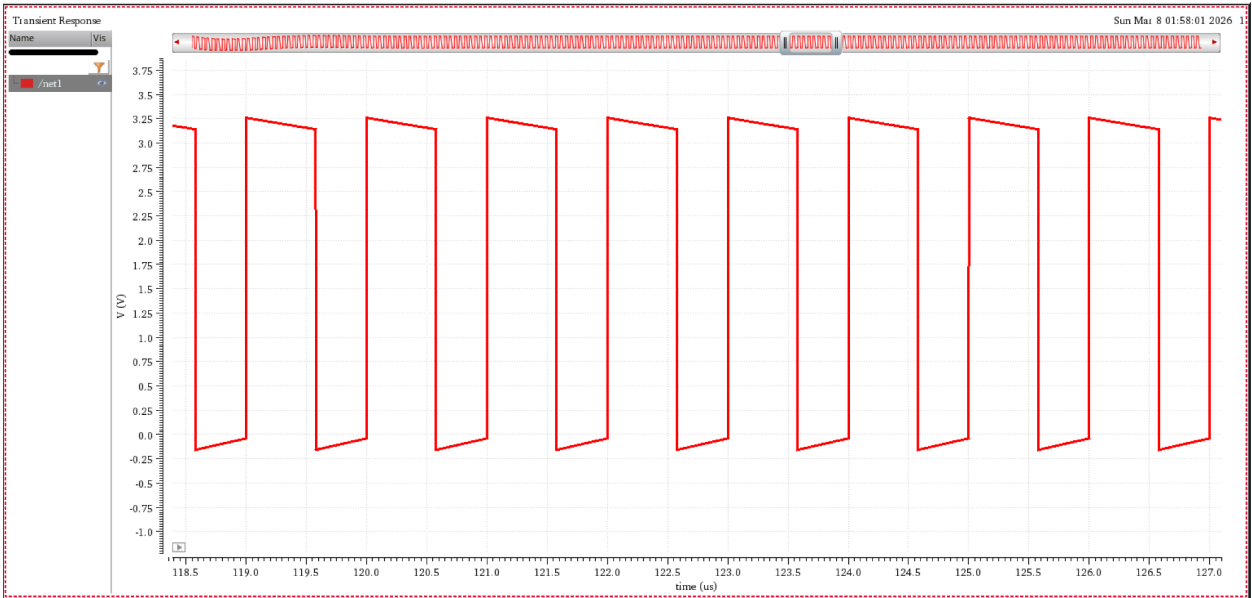
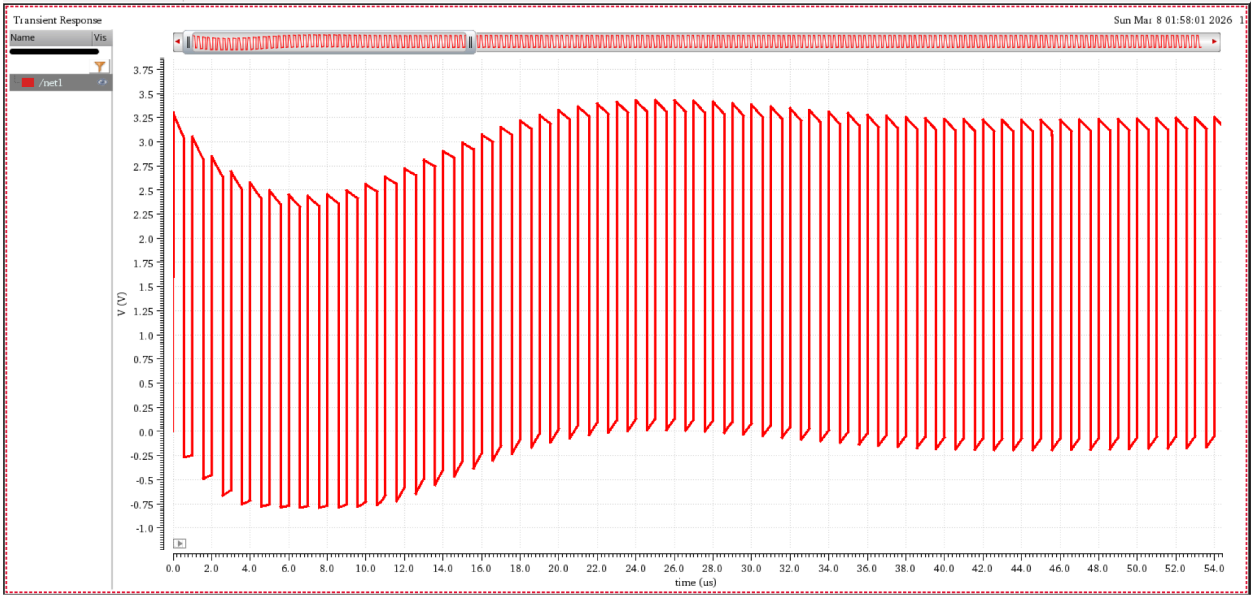
COMBINED PLOT



INDUCTOR CURRENT

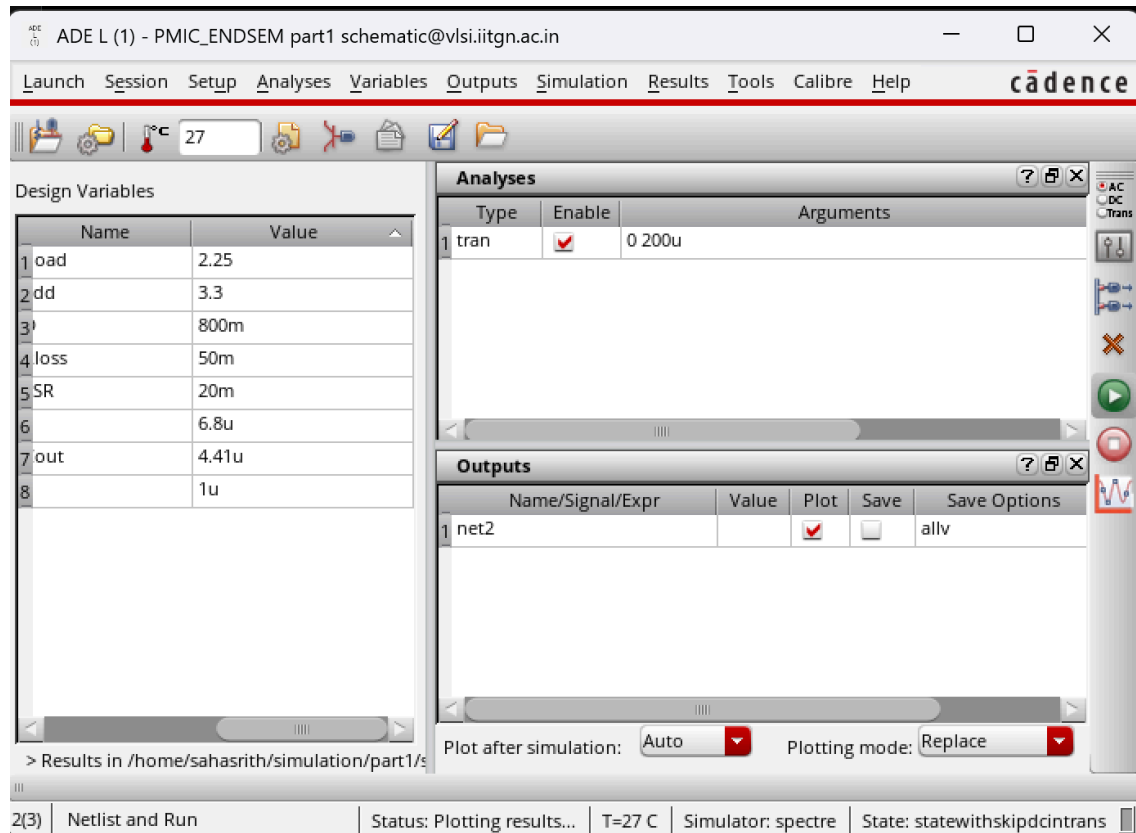


VOLTAGE SWITCHING

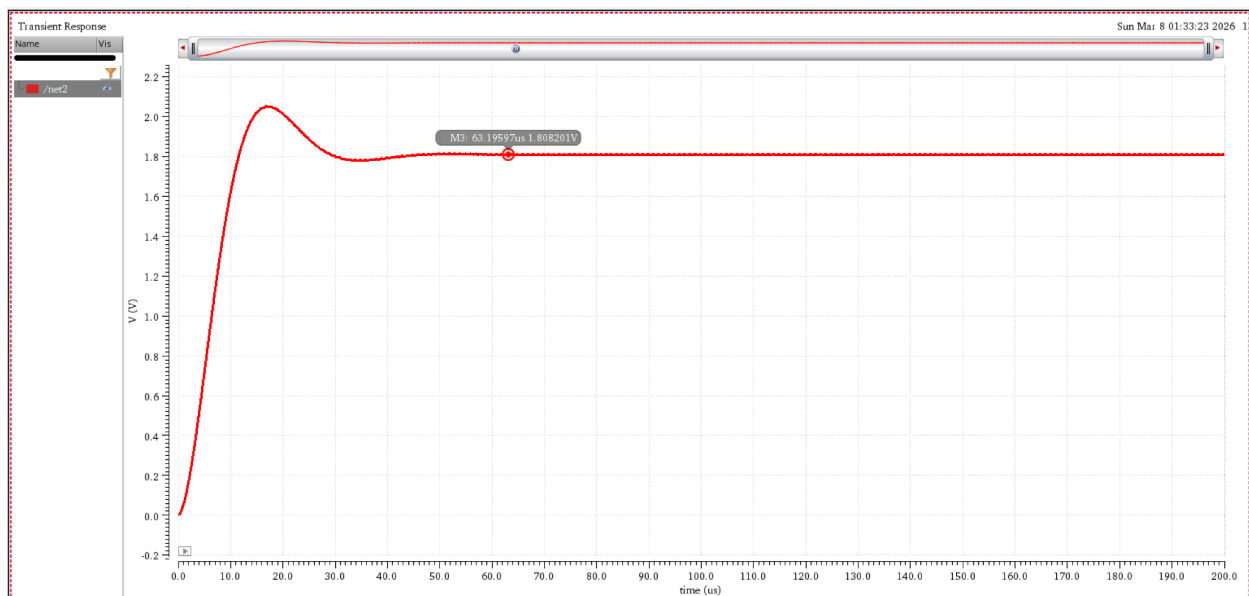


2. For heavy load

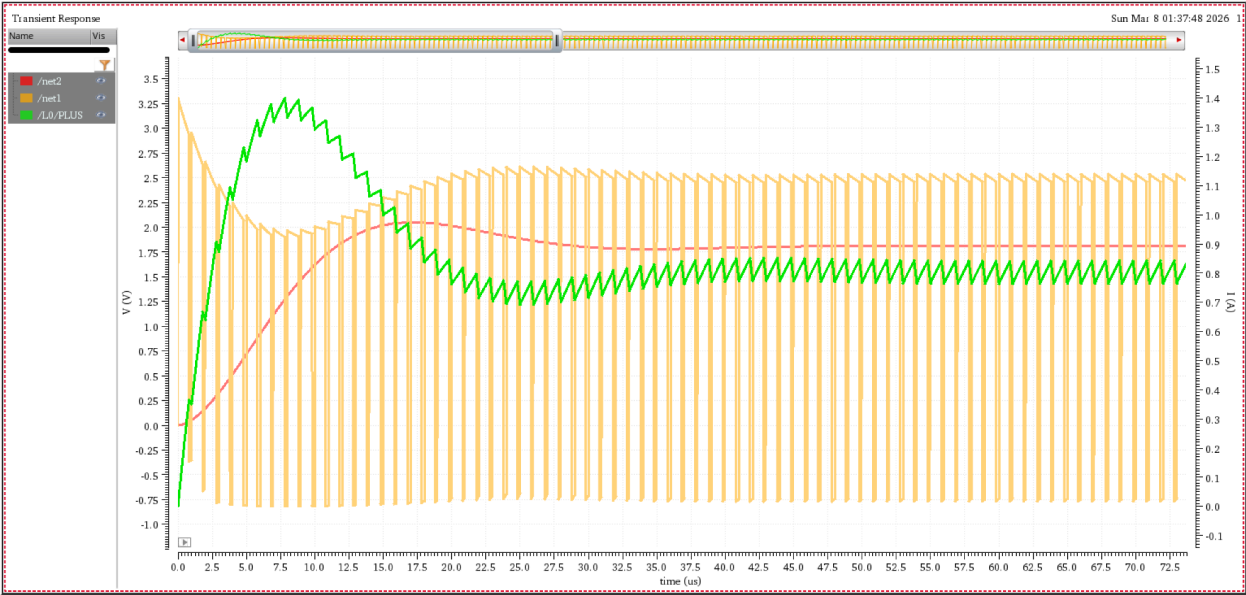
ADL



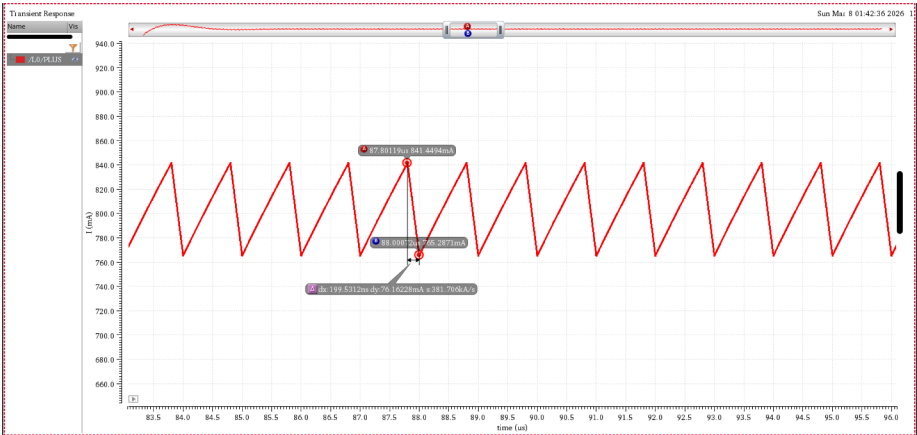
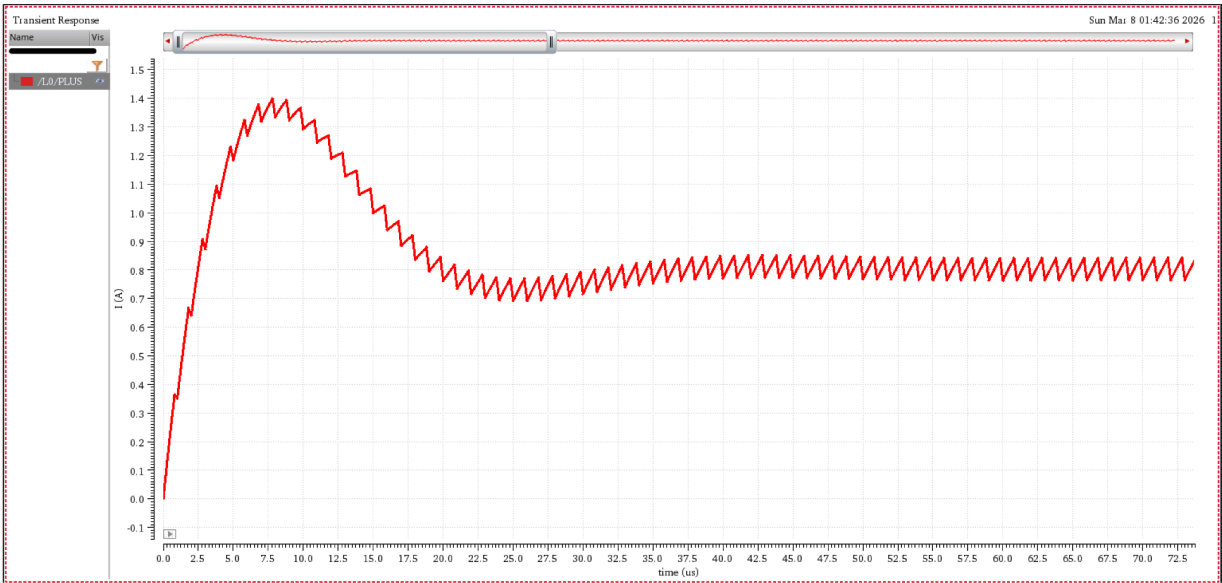
OUTPUT VOLTAGE



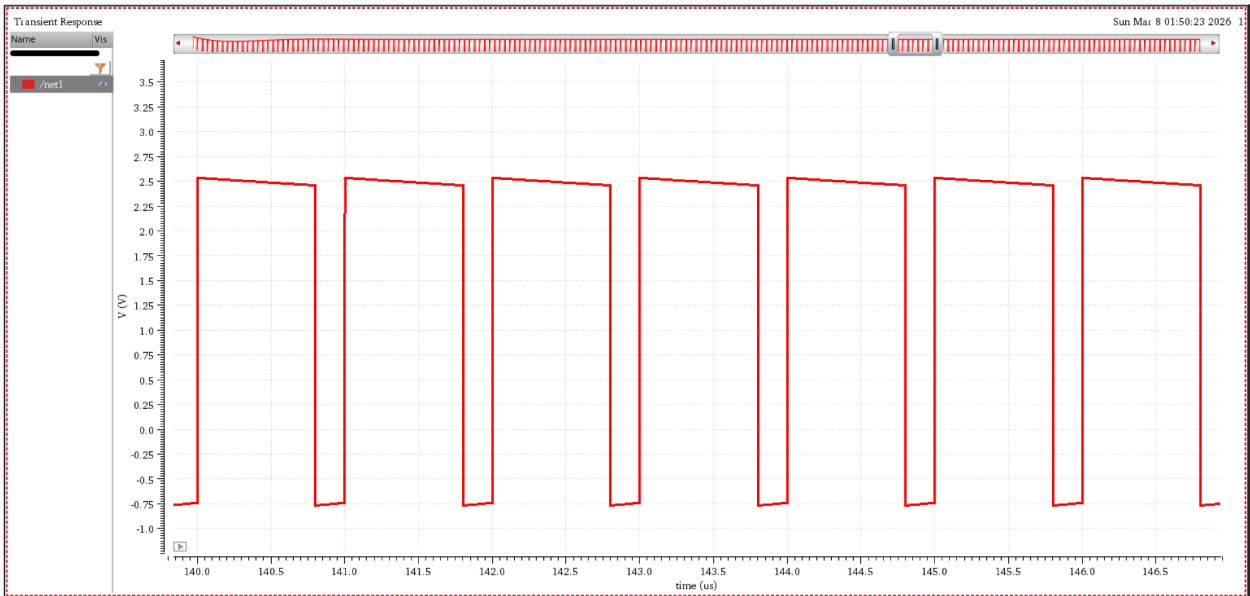
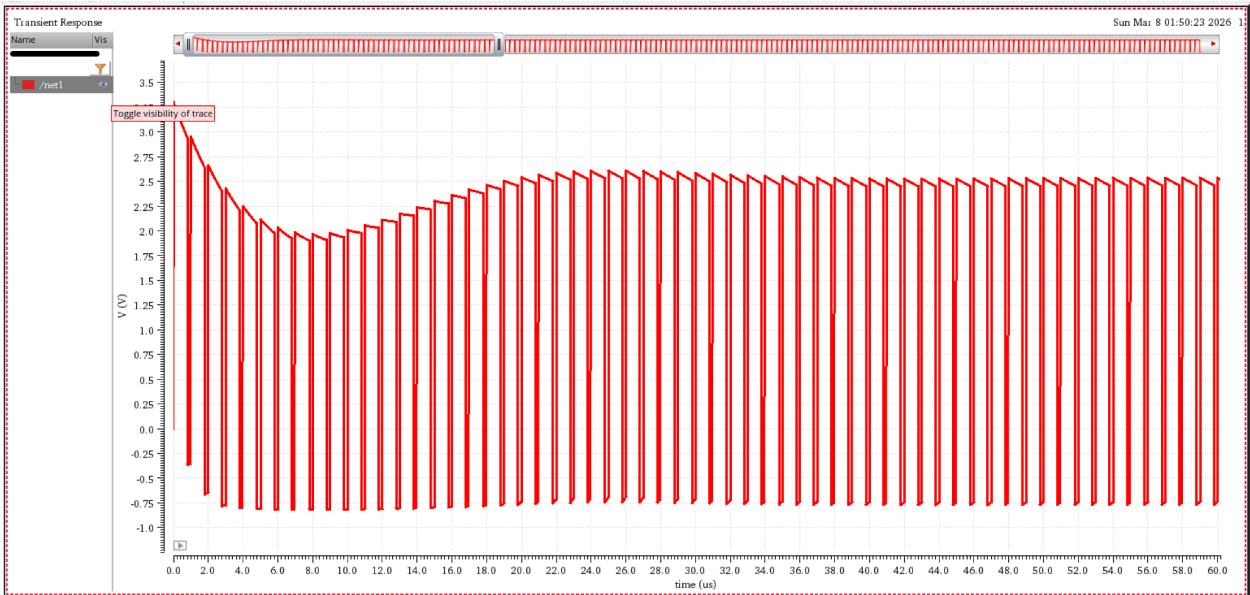
COMBINED PLOTS



INDUCTOR CURRENT



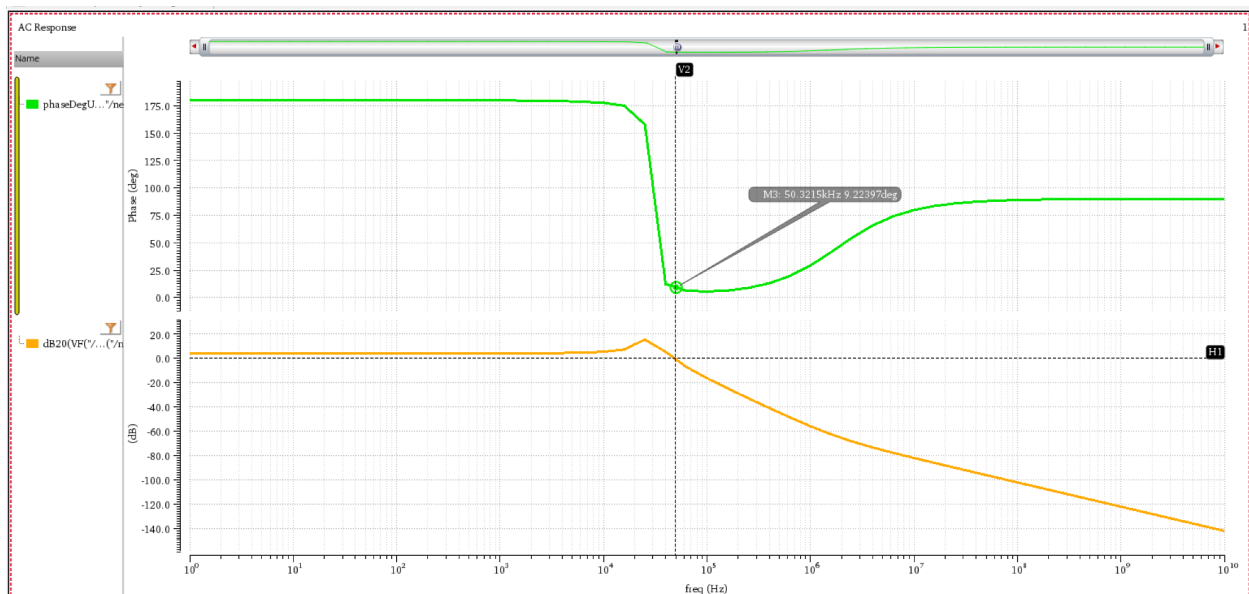
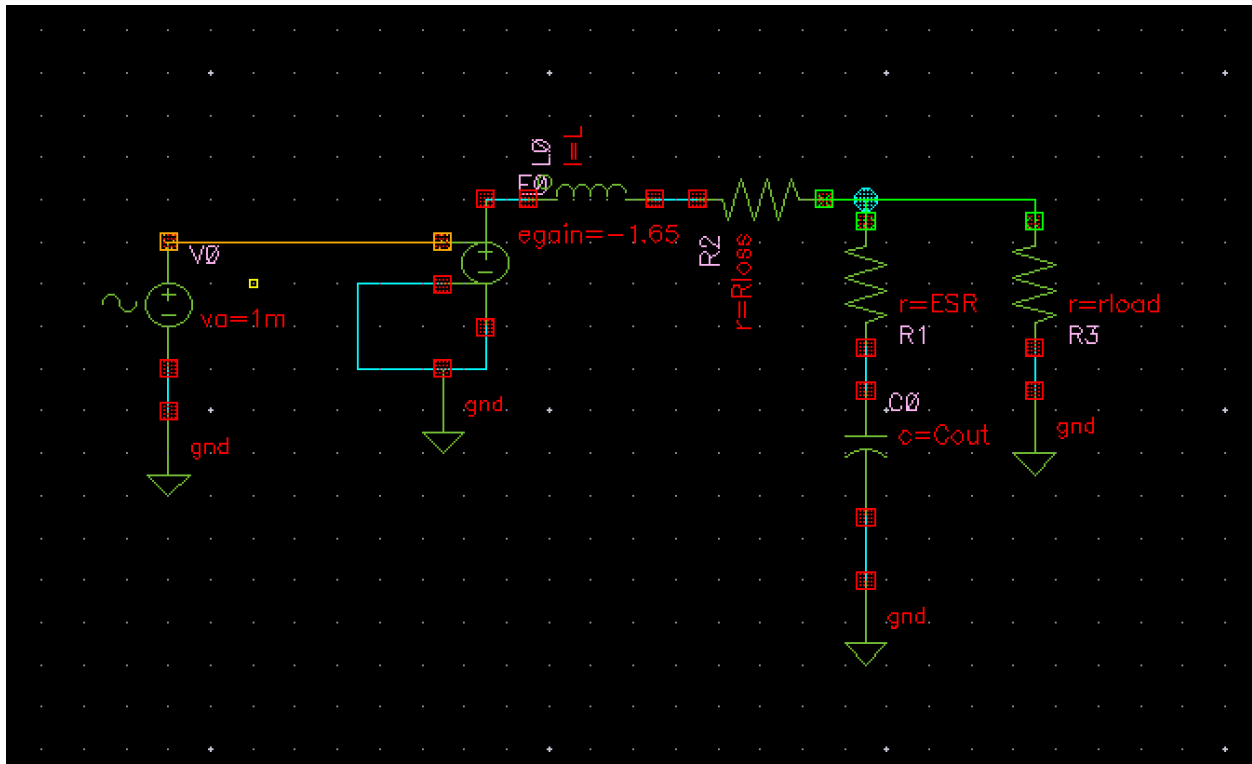
SWITCHING VOLTAGE



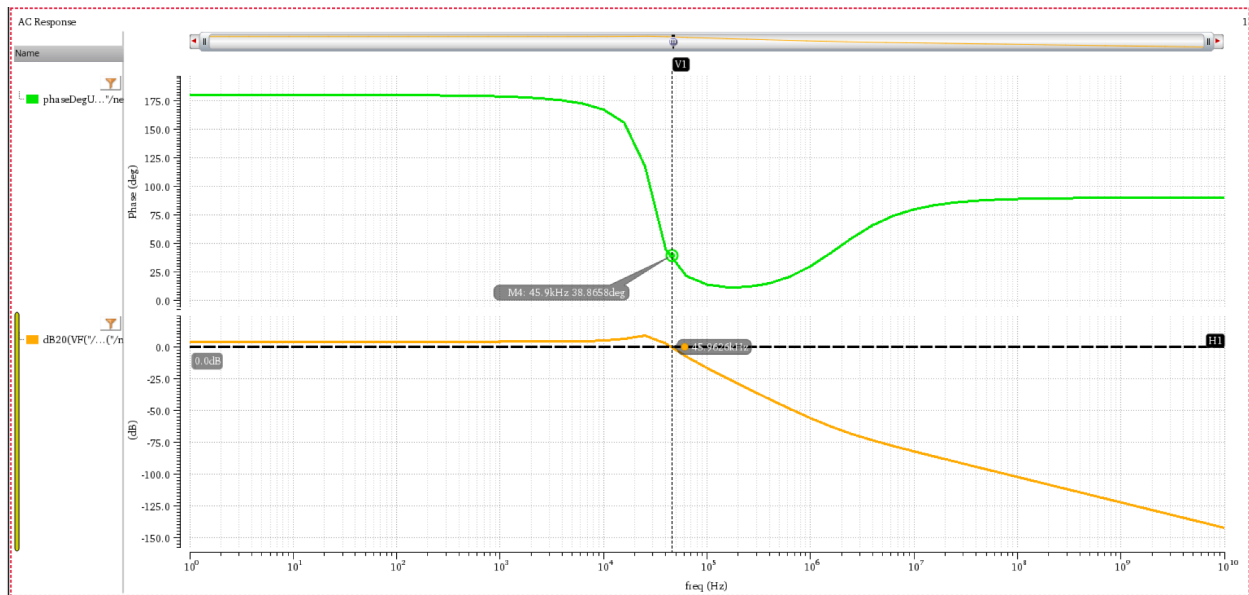
PART - 1B

A. Here i have added gain block which represents the error amplifier and performing the ac analysis

1. Light load



2. Heavy load



So here we have plotted the transfer function of the power stage of SWR

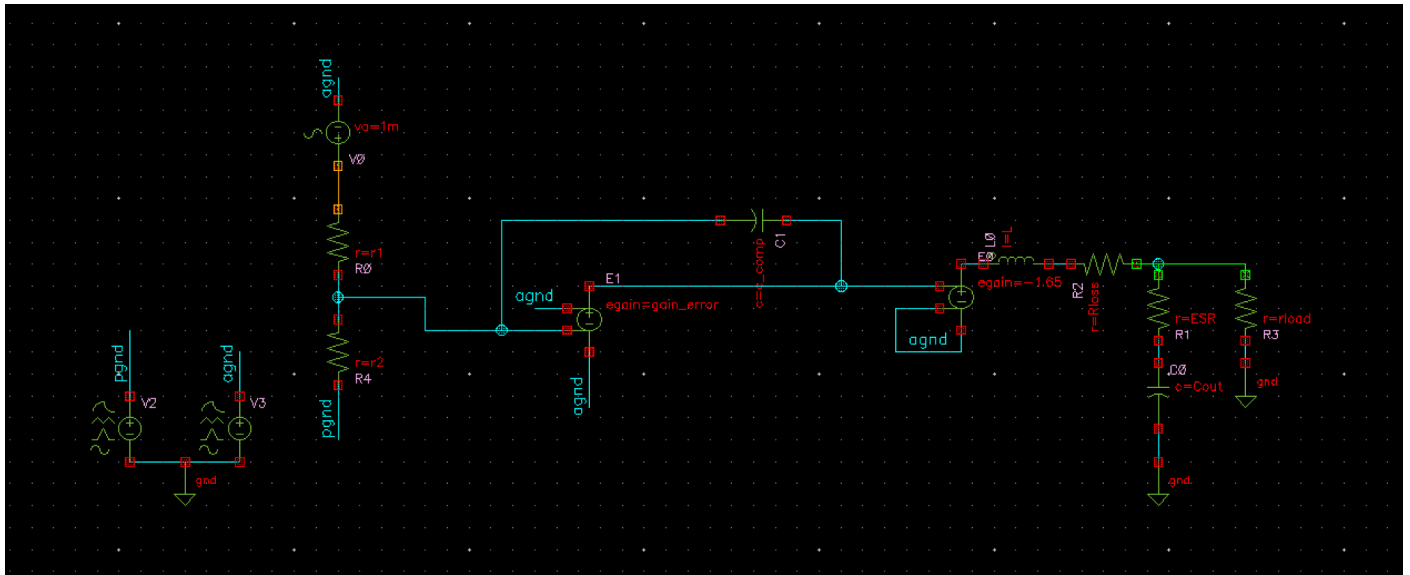
1. Light load PM = 9 degrees
2. Heavy load PM = 38 degrees

From the Bode plot of the power stage transfer function, the system exhibits very low phase margin, especially under light load conditions ($\approx 9^\circ$) and moderate phase margin under heavy load ($\approx 38^\circ$). Such low phase margins indicate that the system is poorly damped and close to instability, which can lead to oscillations and slow transient response in closed-loop operation.

To stabilize the converter and ensure proper regulation, Type-I compensation is employed. Type-I compensation introduces an integrator (pole at the origin) which significantly increases the low-frequency gain, thereby reducing steady-state error and improving DC regulation of the output voltage. Additionally, it helps shape the loop gain so that the crossover frequency occurs in a stable region, improving the overall phase margin and loop stability.

TYPE 1 compensation

Schematic



1. Light load

ADL

ADE L (2) - PMIC_ENDSEM part1B_with_whole_feedback schematic@vlsi.iitgn.ac.in

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

Design Variables

	Name	Value
1	c_comp	5.6n
2	gain_error	1K
3	r1	20K
4	r2	40K
5	rload	18
6	vdd	3.3
7	D	576m
8	Rloss	50m
9	ESR	20m
10	L	6.8u
11	Cout	4.41u
12	T	1u

Analyses

Type	Enable	Arguments
1 tran	☐	0 200u
2 ac	☒	1 10G Automatic Start-St...

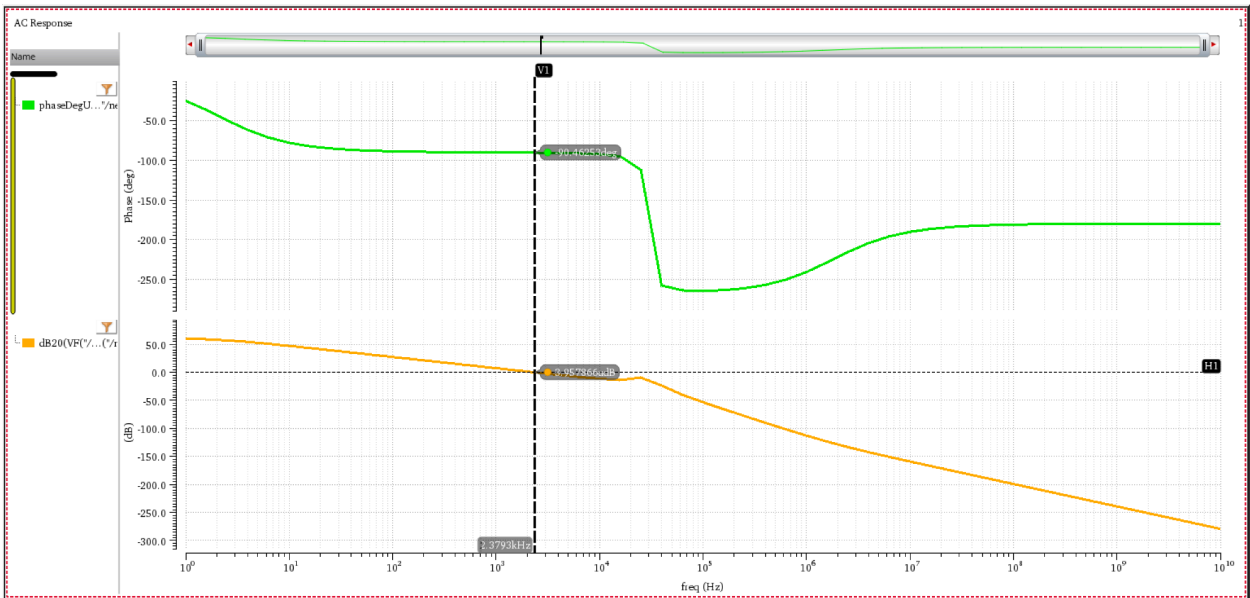
Outputs

Name/Signal/Exp/Alu	Plot	Save	ave Option
1 net2	☐	☐	allv
2 LO/PLUS	☐	☒	yes
3 net1	☐	☐	allv

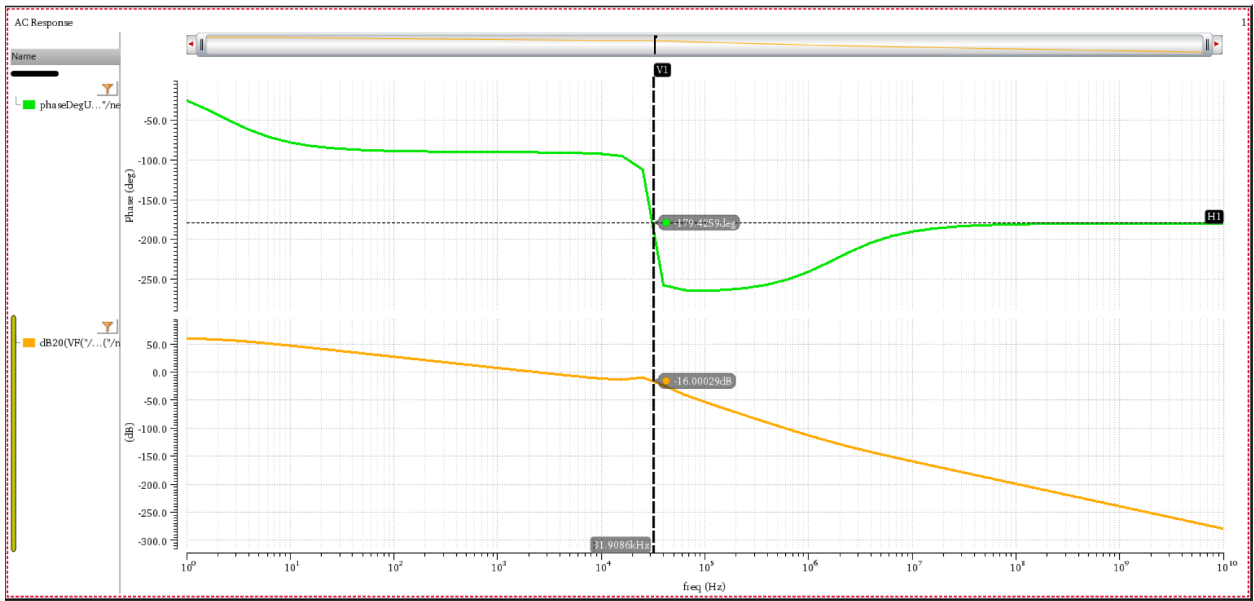
Plot after simulation: Auto Plotting

4(9) Choose Design ... Status: Ready T=27 C Simulator: spectre State: lightload

PHASE MARGIN



GAIN MARGIN



According to plots

PM = 89.54

GM = 16.17

Expression	Value	Expression	Value	Expression	Value
unityGainFreq((VF("/net2") / VF("/ne...	2.404E3	phaseMargin((V...	89.54	gainMargin((VF(...	-16.17

2. Heavy load

ADL

ADE L (2) - PMIC_ENDSEM part1B_with_whole_feedback schematic@vlsi.iitgn.ac.in

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

27

Design Variables

Name	Value
1 c_comp	5.6n
2 gain_error	1K
3 r1	20K
4 r2	40K
5 rload	2.25
6 vdd	3.3
7 D	800m
8 Rloss	50m
9 ESR	20m
10 L	6.8u
11 Cout	4.41u
12 T	1u

Analyses

Type	Enable	Arguments
1 tran	☐	0 200u
2 ac	☒	1 10G Automatic Start-St...

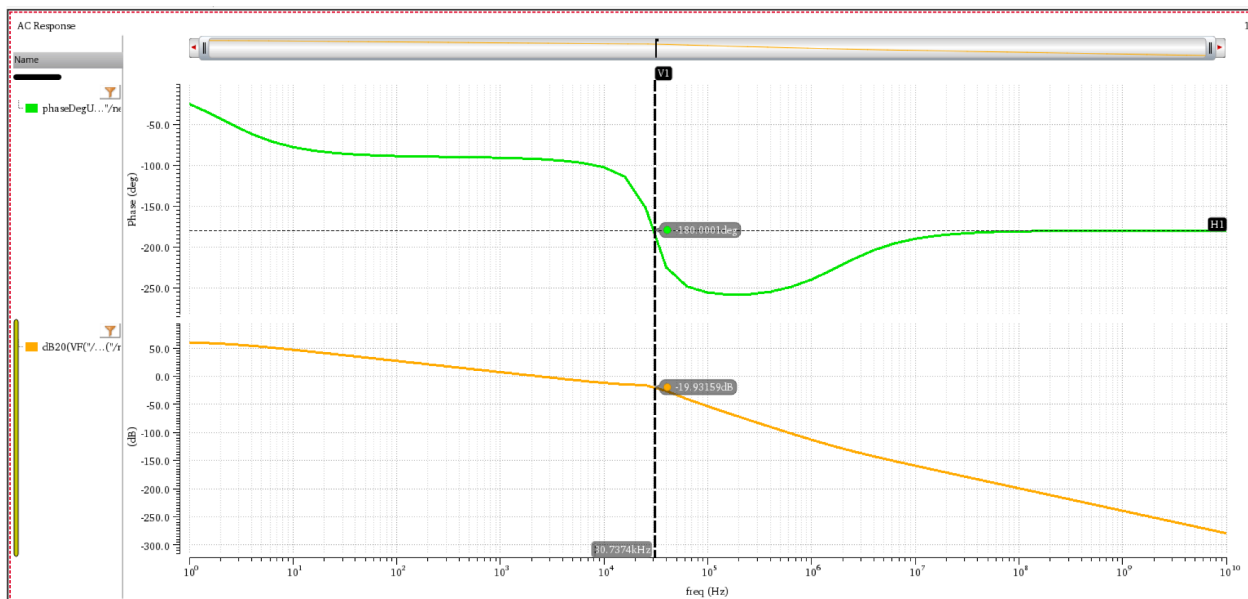
Outputs

Name/Signal/Exp	Plot	Save	Ave Option
1 net2	☐	☐	allv
2 L0/PLUS	☐	☒	yes
3 net1	☐	☐	allv

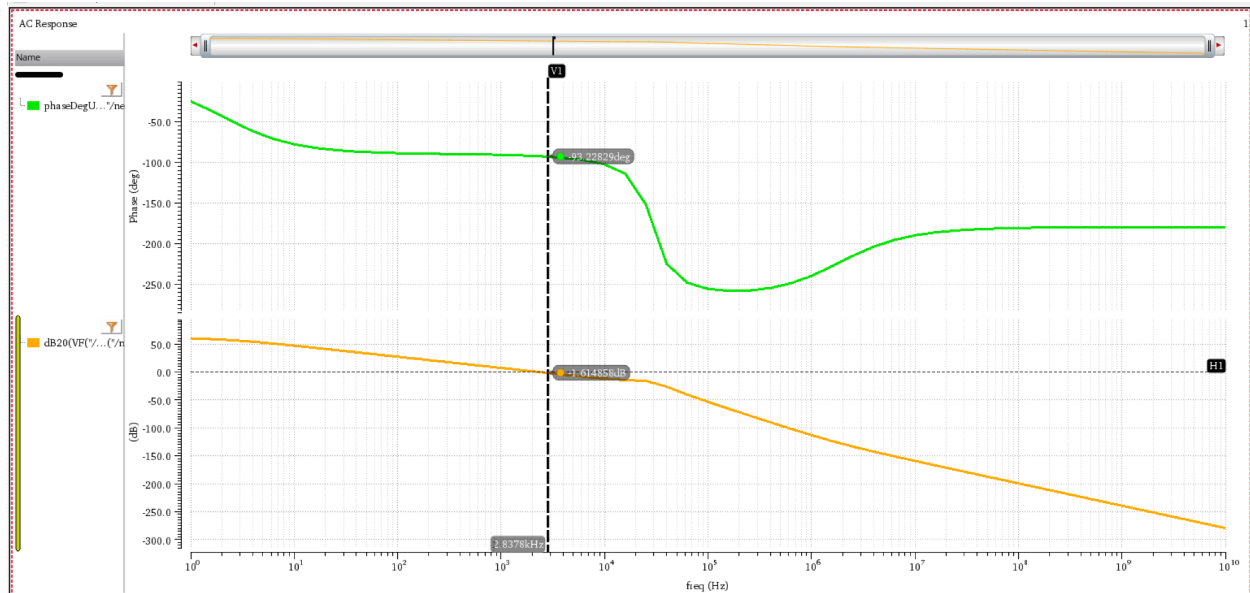
Plot after simulation: Auto Plotting

4(9) Load State ... Status: Ready T=27 C Simulator: spectre State: lightload

GAIN MARGIN



PHASE MARGIN



According to plots

PM = 87.41

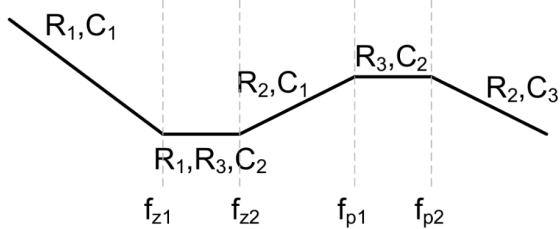
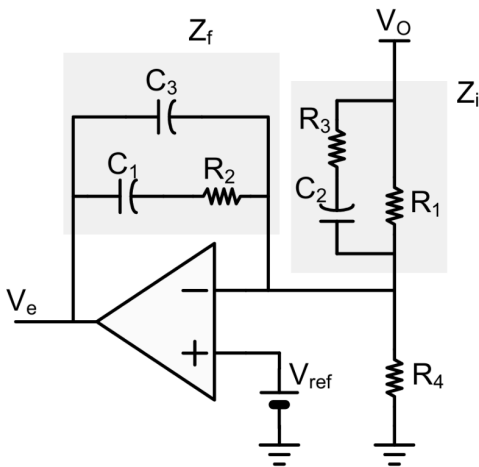
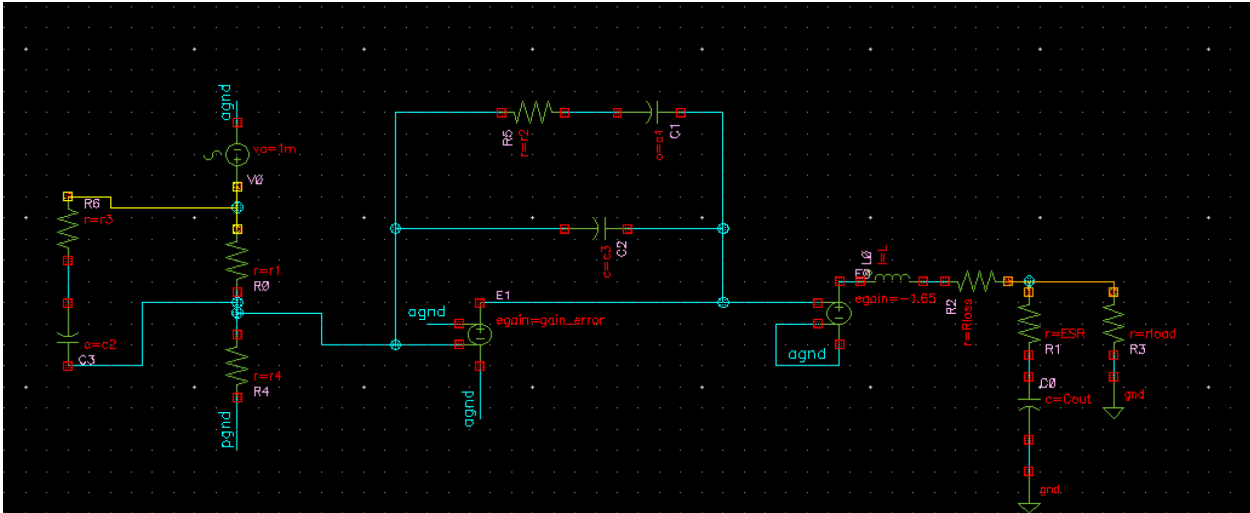
GM = 19.61

phaseMargin(VF('net2') / VF('net...'))					
Expression	Value	Expression	Value	Expression	Value
1 phaseMargin(VF('net2') / VF('net...'))	87.41	gainMargin(VF('net2') / VF('net...'))	-19.61	unityGainFreq(VF('net2') / VF('net...'))	2.368E3

Type-I compensation improves the low-frequency gain and reduces steady-state error, but it results in a very low unity gain bandwidth (UGB). Because of this low UGB, the loop bandwidth remains small, leading to slow transient response and insufficient phase margin improvement. Since the buck converter power stage contains an LC double pole, Type-I compensation cannot effectively compensate for the large phase lag. Therefore, Type-III compensation is used to introduce additional zeros and poles, which increase the UGB, provide phase boost near crossover, and improve overall stability and transient performance.

TYPE 3 compensation

1. Light load



Component	Value
C1	2.7 nF
C2	220 pF
C3	2 pF
R1	20 kΩ
R2	75 kΩ
R3	1.1 kΩ
R4	40 kΩ

ADE L (4) - PMIC_ENDSEM part1B_type3_comp_with_whole_feedback schematic@vlsi.iitgn.ac.in

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

27

Design Variables

Name	Value
1 c1	2.7n
2 c2	220p
3 c3	2p
4 r1	20K
5 r2	75K
6 r3	1.1K
7 r4	40K
8 gain_error	1K
9 rload	18
10 vdd	3.3
11 D	576m
12 Rloss	50m
13 ESR	20m
14 L	6.8u
15 Cout	4.41u
16 T	1u

>

Analyses

Type	Enable	Arguments
1 tran	☐	0 200u
2 ac	☒	1 100G Automatic Start-Stop

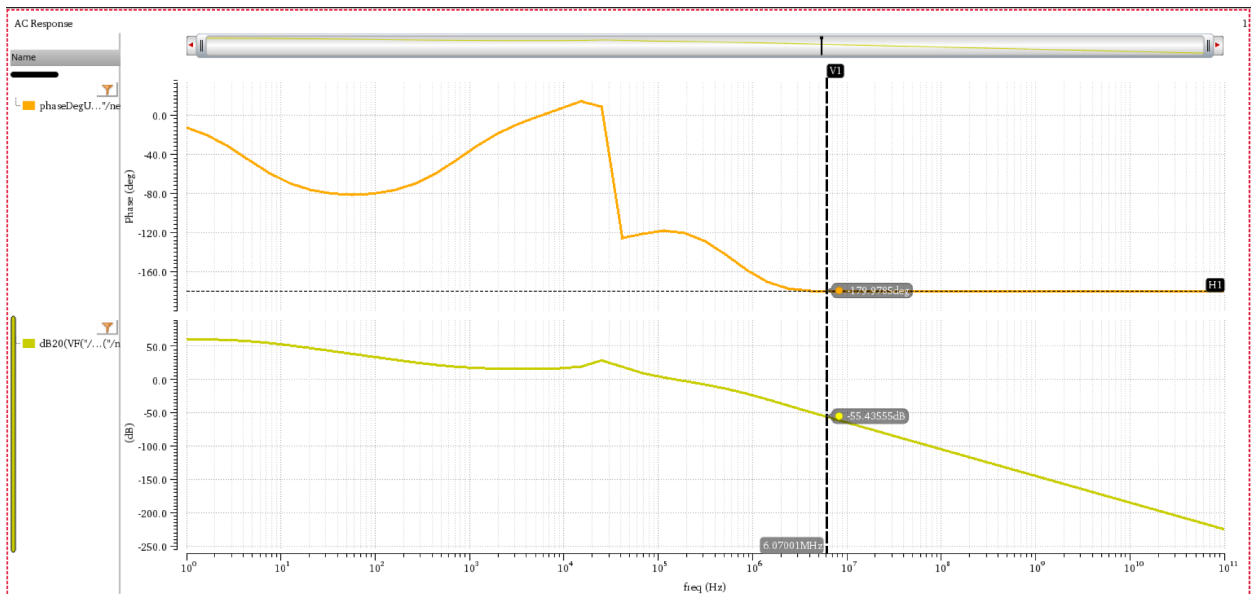
Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 net2		☐	☐	allv
2 LO/PLUS		☐	☒	yes
3 net1		☐	☐	allv

Plot after simulation: Auto Plotting mode: New Win

19(63) AC Gain & Phase Status: Ready T=27 C Simulator: spectre State: type3complightload_problemwithPM60.6_GM55

GAIN MARGIN



2. Heavy load

ADL

ACADE L (4) - PMIC_ENDSEM part1B_type3_comp_with_whole_feedback schematic@vlsi.iitgn.ac.in

Launch Session Setup Analyses Variables Outputs Simulation Results Tools Calibre Help

Design Variables

Name	Value
c1	2.7n
c2	220p
c3	2p
r1	20K
r2	75K
r3	1.5K
r4	40K
gain_error	1K
rload	2.25
vdd	3.3
D	800m
Rloss	50m
ESR	20m
L	6.8u
Cout	4.41u
T	1u

Analyses

Type	Enable	Arguments
tran	☐	0 200u
ac	☒	1 100G Automatic Start-Stop

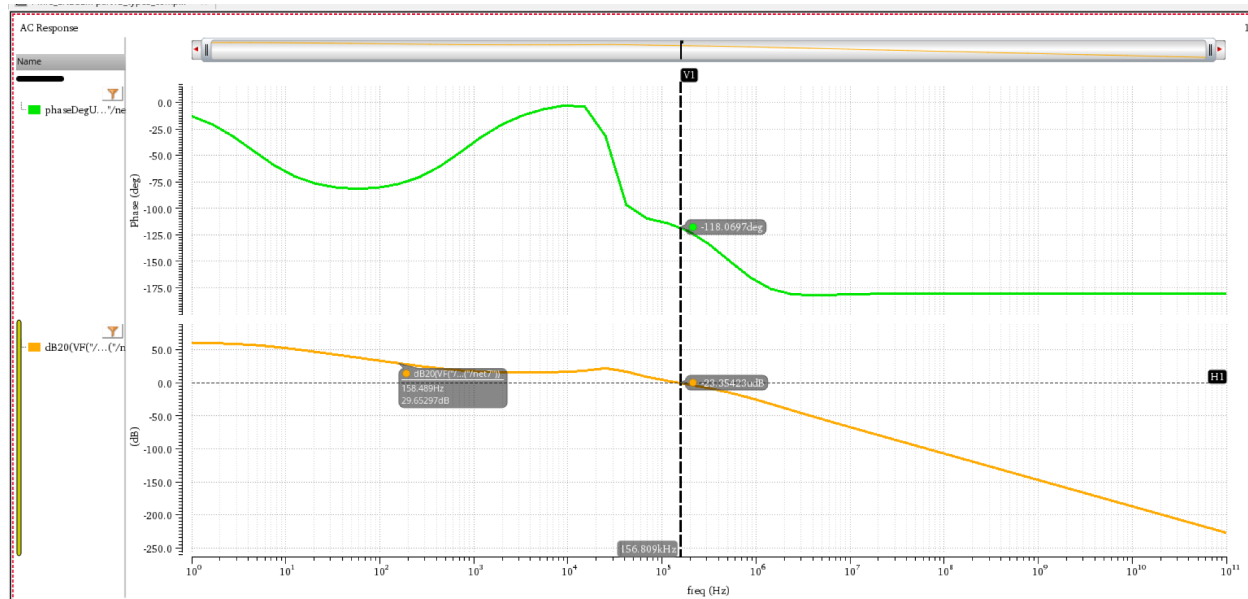
Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
net2		☐	☐	allv
L0/PLUS		☐	☒	yes
net1		☐	☐	allv

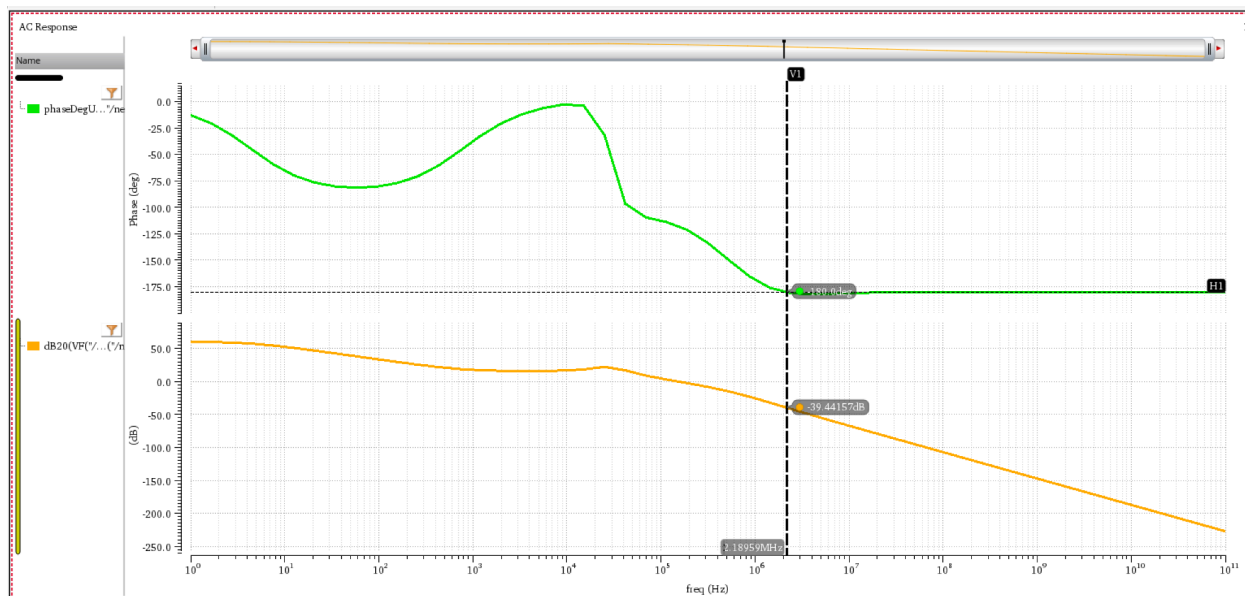
Plot after simulation: Auto Plotting mode: New Win

19(63) Netlist and Run Status: Ready T=27 C Simulator: spectre State: type3compheavyload_problemwithPM62GM32

PHASE MARGIN



GAIN MARGIN



The major benefit of Type-III compensation over Type-I compensation is that Type-III provides additional phase boost using two zeros, which allows the loop to achieve a higher unity gain bandwidth (UGB) while maintaining sufficient phase margin. In contrast, Type-I compensation only has an integrator and no phase boost, so the UGB must be kept very low (well below the LC double pole) to maintain stability. Therefore, Type-III compensation is used in switching converters to achieve higher bandwidth and stable operation.

